

Sum of four square

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Claim : Every non negative integer can be represent by at most 4 square number.

Proof : First we need to prove that if two given distinct prime number p_1 and p_2 is fulfill the claim then they the product of them also fulfill the claim and since according to the fundamental of arithmetic every number greater than 1 can be represent as a series of product of prime number

The first part is prove by Euler,we can just assume that $p_1 = \sum_{i=1}^4 x_i^2$ and $p_2 = \sum_{i=1}^4 y_i^2$ then the product can be represent by 4 square number,so we focus on second part which is prime number p can be represent as at most 4 square number

We can prove that $(a^2 + b^2 + 1) \equiv 0 \pmod{p}$, the reason is simple first we change the equation to $a^2 \equiv -1 - b^2 \pmod{p}$, we assume that $0 \leq a, b \leq \frac{p-1}{2}$ then since a and b are coprime to p so $a^2 \pmod{p}$ and $-1 - b^2 \pmod{p}$ will produce exactly $\frac{p+1}{2}$ result and the total result is $\frac{p+1}{2} + \frac{p+1}{2} = p + 1$, so according to pigeon principle, it must have at least one element exist both side, so at the end we prove that

$$m \cdot p = a^2 + b^2 + 1$$

and since we assume $0 \leq a, b \leq \frac{p-1}{2}$ then

$$m \cdot p = \left(\frac{p-1}{2}\right)^2 + \left(\frac{p-1}{2}\right)^2 + 1 < \left(\frac{p}{2}\right)^2 + \left(\frac{p}{2}\right)^2 + 1$$

$$m \cdot p < \frac{p^2}{2} + 1$$

$$m < \frac{p}{2} + \frac{1}{p}$$

$$m < p$$

now we know $0 \leq m < p$ but m may not be one, so the proof still not end, we can construct a number by using $y_i \equiv x_i \pmod{m}$, x_i is one of the four square number to construct $m \cdot p$, now consider $\sum y_i$

$$y_1^2 + y_2^2 + y_3^2 + y_4^2 \equiv x_1^2 + x_2^2 + x_3^2 + x_4^2 \pmod{m}$$

$$y_1^2 + y_2^2 + y_3^2 + y_4^2 \equiv 0 \pmod{m}$$

then let $m \cdot m_1 = y_1^2 + y_2^2 + y_3^2 + y_4^2$, since $y_i \equiv x_i \pmod{p}$, we can change the range of y_i from $[0, m-1]$ to $(-\frac{m}{2}, \frac{m}{2}]$, so after changing the range we can observe that only when all $y_i = \frac{m}{2}$, $m \cdot m_1$ will be the largest which is equal to m^2 but it is impossible because from the previous result we know $m \cdot p \not\equiv 0 \pmod{m^2}$, if $y_i = \frac{m}{2}$ will cause $m \cdot p \equiv 0 \pmod{m^2}$ which is contradiction

$$\begin{aligned}
y_i &\equiv x_i \pmod{m} \\
x_i &= k \cdot m + y_i \\
x_i^2 &= k^2 \cdot m^2 + 2 \cdot k \cdot m \cdot y_i + y_i^2 \\
x_i^2 &= k^2 \cdot m^2 + 2 \cdot k \cdot m \cdot \frac{m}{2} + y_i^2 \pmod{m^2} \\
x_i^2 &\equiv y_i^2 \pmod{p} \\
\sum x_i^2 &\equiv \sum y_i^2 \pmod{p} \\
m \cdot p &\equiv 0 \pmod{p}
\end{aligned}$$

so since not all $y_i = \frac{m}{2}$ then

$$\begin{aligned}
m \cdot m_1 &< m^2 \\
m_1 &< m
\end{aligned}$$

at the end we let $m \cdot p$ multiply $m \cdot m_1$, according to the proof by Euler, we know the result also a four square number

$$\begin{aligned}
m \cdot p \cdot m \cdot m_1 &= (\sum x_i) \cdot (\sum y_i) \\
m_1 \cdot p &= \frac{1}{m^2} \cdot (\sum x_i) \cdot (\sum y_i)
\end{aligned}$$

obviously $m_1 \cdot p < m \cdot p$, so we can keep repeat this operation until $m = 1$, proved.